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Upcoming course
ML, DL, NLP

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~~Multiple~~ Linear Regression
understanding

lec - 6 Multiple Linear
regression

lec - 7 Performance Metrics

lec - 8 MSE, MAE, RMSE

In a Dataset, the multiple
Linear Regression, when,
there're more than one
inputs.

$$\text{Model} = \theta_0 + \theta_1 x \rightarrow \text{If } x \text{ is Independent}$$

θ_0 = Intercept

θ_1 = Slope

→ Suppose we have a dataset
of House pricing dataset

Real dataset

can be presented as

Independent Features			
No. of Rooms	Size of House	Location	Price

$$y = \theta_0 + \theta_1 x_1 +$$

$$\theta_2 x_2 + \theta_3 x_3$$

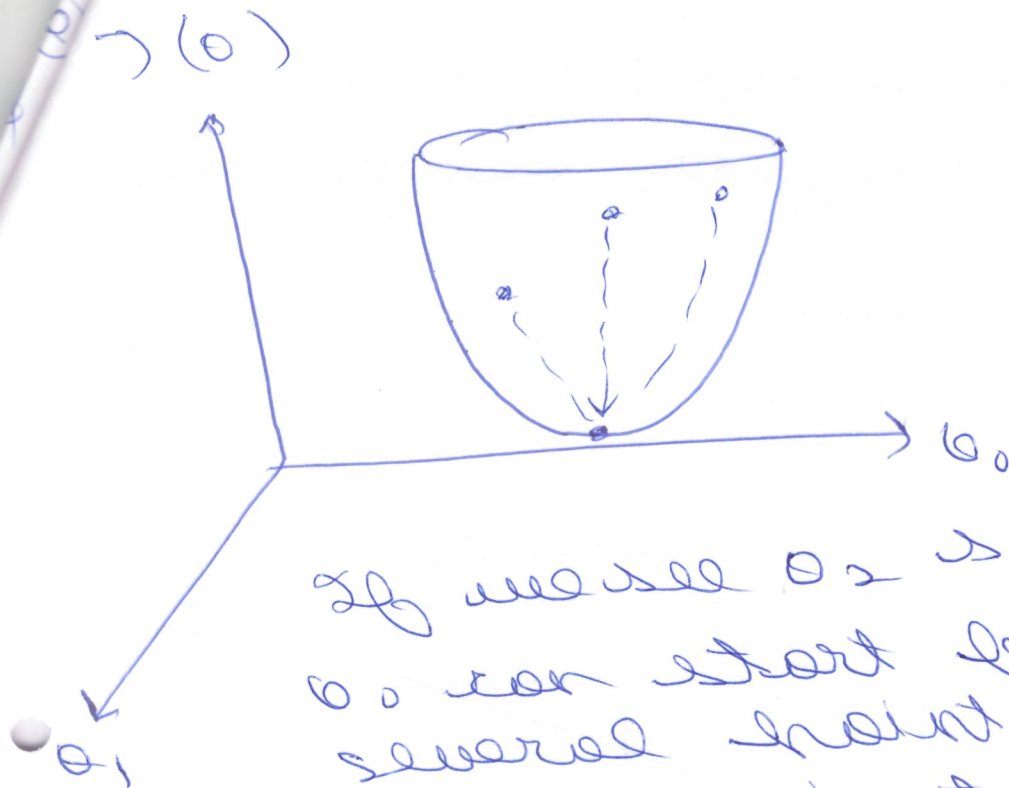
{ Multiple Linear Regression

Output feature

→ Here we are showing 3 coefficients

Here, $\theta_1, \theta_2, \theta_3$ = coefficients

θ_0 = Intercept



If we see $b_2 > b_3$,
 b_0 can start from
 several points.

But the aim is to converge
 to this global minimum.

$$J(b) = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3$$

more Input features

=> Performance metrics
used in Linear Regression

(1) R squared

(2) ~~Adjusted~~
test $R =$
squared

$$R_{\text{squared}} = 1 - \frac{SS_{\text{res}}}{SS_{\text{total}}}$$



this is further written
as:

$$= 1 - \sum (y_i - \hat{y}_i)^2$$

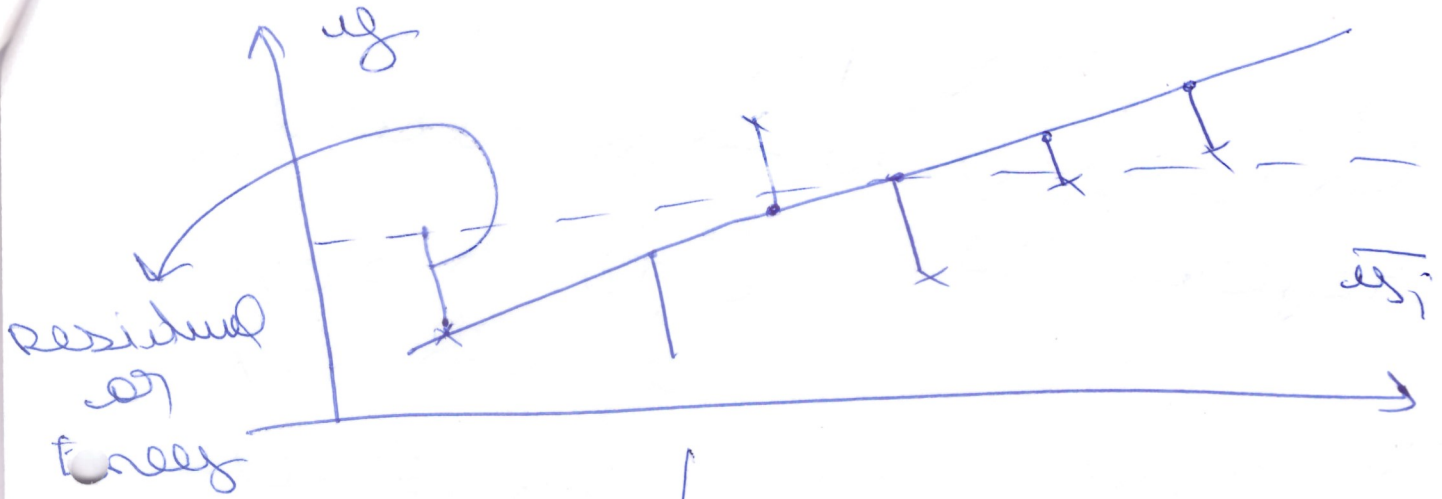
↳ summation of
differences in y_i &
 $\hat{y}_i$

these are said to be as
residual errors

(5)

$$R_{\text{square}} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

we just fit line graph.



This is the average
of the true points.

→ use as the average
of all the y_i

So, basically, in detail,
it can be written as:-

$$R_{\text{square}} = \frac{1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}}{1 - \frac{\sum (y_i - \bar{y})^2}{\sum (y_i - \bar{y})^2}}$$

$$= 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

$$= 1 - \frac{\text{Small Number}}{\text{Bigger Number}}$$

$$= 1 - \text{smaller number}$$

≈ 1
 $\hookrightarrow$ Here 0.20 $\rightarrow$ Basically refers to:

$$0.20 \Rightarrow 20\%$$

$$0.85 \Rightarrow 85\%$$

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$$0.90 = 90\%$$

Now, if we have to verify that whether model is underfitting or overfitting.

(2) Adjusted R squared

Suppose, we have a dataset to predict price of the house

Dataset $\rightarrow$ Price

Size of the house

Price of house.

⑦ As the size of house increases, the price is also going to increase

→ There is a direct correlation.

→ This is a positive correlation.

• ~~→ zero~~
⇒ The R^2 squared value is around 73%.

R^2 squared value is around 73% ⇒ 0.73

→ Suppose, we include one more variable.

• Size of house ⇒ No. of bedrooms

• As the no. of bedrooms increases, the price is also going to increase

→ Here also, we can see a +ve correlation.

Hence R squared value will definitely increase

→ It has now increased to 85%.

$$R \text{ squared} \Rightarrow 85\% \Rightarrow 0.85$$

⇒ Adjusted R squared

$$= 1 - \frac{(1 - R^2)(N - 1)}{N - p - 1}$$

N = No. of data points

p = No. of independent features that

we're using.

→ N = No. of data points

p = No. of Independent variables.

Suppose, initially $p = 2$

$$R^2 = 90\%$$

→ If we apply same formula with R^2 adjusted

$$R^2 \text{ adjusted} = 80\%$$

$$p = 3$$

$$R^2 = 92\%$$

$$q = 3 \quad R^2 = 0.27$$

(9)

→ Hence R^2 adjusted value will get reduced.

→ It would be less than the previous one

⊛ This scenario is seen when R^2 adjusted is lower

If the additional independent feature is directly proportional to output feature

MSE, MAE, RMSE
[Cost Fun]

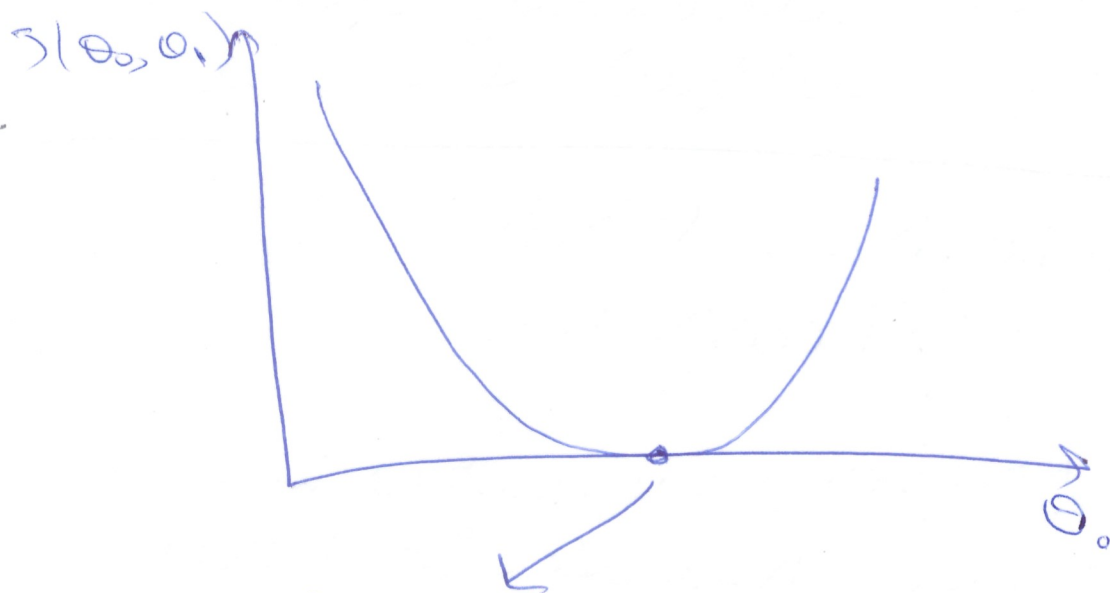
• MSE $\Rightarrow$ Mean Squared Error

MAE $\Rightarrow$ Mean Absolute Error

RMSE $\Rightarrow$ Root Mean Squared Error

[Cost function] $\rightarrow$ Performance metrics

Suppose we have a data set of salary vs hours of experience



Basically, here we have to come to this global minimum.

Experience	Salary
-	-
-	-
-	-

(11)

Our main aim is to reduce the error

The first type of cost function mainly used:

(1) Mean Squared Error:

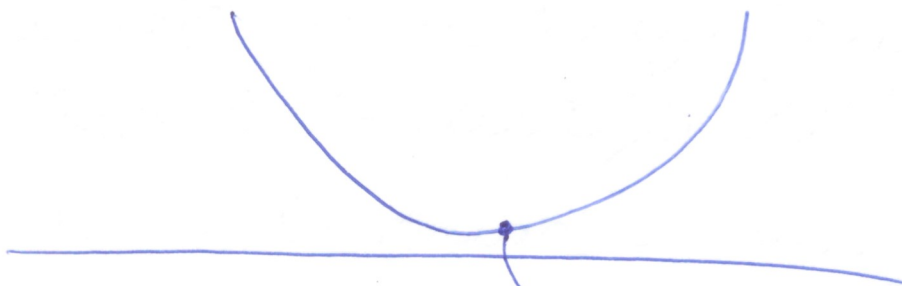
$$MSE = \sum_{i=1}^n \frac{(y - \hat{y})^2}{n}$$

The best fit line is created in such a way that cost function must decrease.

$$y = a + bx + c$$

→ The quadratic equation can be written as:

We will be getting a gradient descent



Global minimum
error will be less

The advantages of
using least squares
Error include:-

~~Advantages~~ (1) They're differentiable
& slope can be calculated
at every point.

(2) It has just one local
& one global minimum

Disadvantages:-

(1) Not robust to outliers

(2) when we try to
calculate:-

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Here, we find differences b/w
actual & predicted values.
It's no longer in same
unit.

Error $\Rightarrow 2.5 \Rightarrow$ (label 2)

It is difficult to understand, when we compare with truth value

$\Rightarrow$ Mean Absolute Error

MAE is given by the formula as

$$MAE = \frac{1}{n} \sum_{i=1}^n |y - \hat{y}|$$

$\rightarrow$ Formula

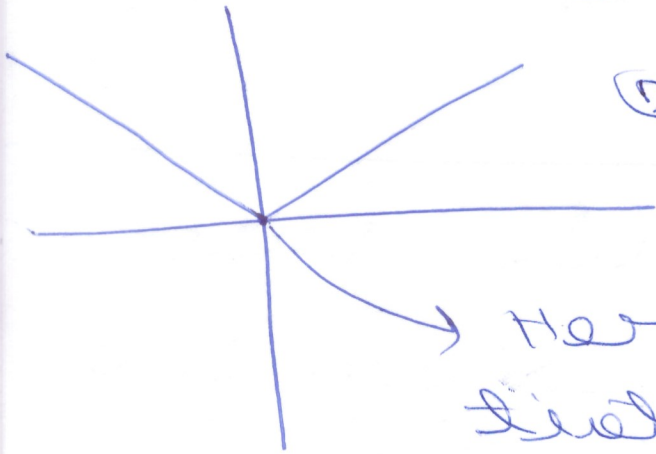
$$MAE = \frac{1}{n} \sum_{i=1}^n |y - \hat{y}|$$

Advantage:-

- ① This will be robust to outliers.
- ② It will be in the same unit.

$\Rightarrow$ Disadvantages :-

① Convergence also usually takes more . optimization is a complex task.



② Time consuming

Here we can't differentiate to find any value at

Here, we create the sub gradients.

(3) RMSE (Root Mean Squared Error)

$$RMSE = \sqrt{MSE}$$

$$= \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

$\rightarrow$ This is the entire formula of RMSE.

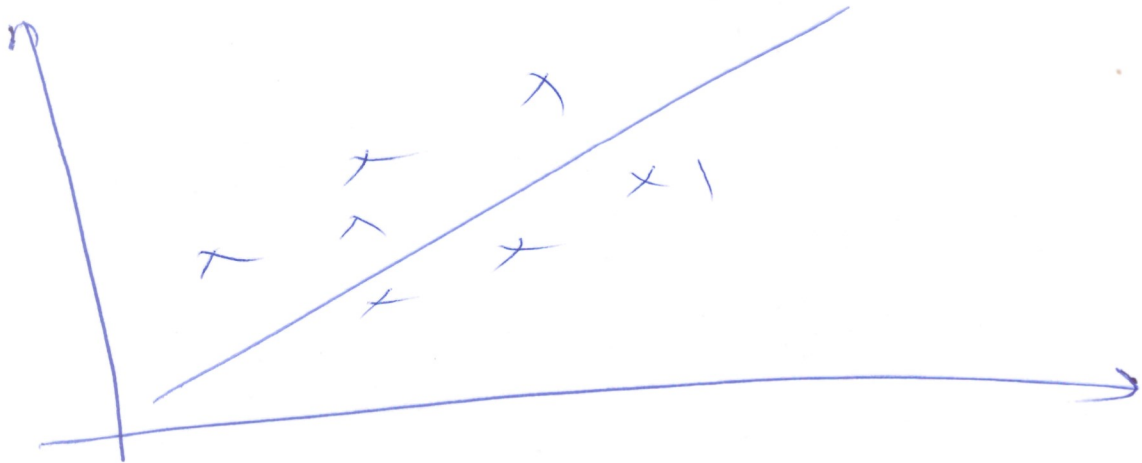
Advantages & Disadvantages 15

Advantages

- (1) Same Unit
- (2) Differentiable.

Disadvantages

- (1) Not robust to outliers
- (2) We must look at all 3 of them.



$\rightarrow$ n SE
 R^2 SE
 n SE
 R^2
 Adjusted R^2

$\rightarrow$ These all form performance metrics